Project Demo Planning

Date	30 June 2026
Team ID	
Project Name	VBCUA
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan f o r demonstrating the project, including the f low of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	Introduction	Introduce the team, project objectives, and the problem of evaluating conceptual understanding through spoken explanations.	2	Member 1
2	Solution Overview	Explain the VBCUA workflow, system architecture, AI pipeline, and technologies used (Whisper, Sentence-BERT, Librosa, Streamlit).	3	Member 2
3	Frontend Demonstration	Demonstrate the Streamlit interface, topic selection, audio upload, and application features.	3	Member 3
4	Live Analysis	Upload a sample audio file and demonstrate speech transcription, semantic similarity evaluation, speech metrics, final score, waveform visualization, and PDF report generation.	4	Member 4
5	Backend & AI Integration	Explain how app.py integrates with speech_to_text.py, semantic_eval.py, audio_utils.py, scoring_engine.py, and report_generator.py to perform end-to-end analysis.	3	Member 2
6	Conclusion & Q&A	Summarize the project, discuss future enhancements (multilingual support, user authentication, cloud deployment, progress tracking), and answer audience questions.	5	All Members

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Introduce the project, explain the challenges of evaluating conceptual understanding through spoken explanations, and state the project objectives.
2	Solution Overview	Present the system architecture, application workflow, and technologies used, including Streamlit, OpenAI Whisper, Sentence-BERT, Librosa, PyTorch, and ReportLab.
3	Live Feature Demonstration	Upload a sample audio file, demonstrate speech-to-text transcription, semantic similarity analysis, speech metrics (RMS energy, pause ratio, filler ratio), final understanding score, waveform visualization, and PDF report generation.
4	Q&A Session	Answer evaluator questions, explain the AI pipeline and project modules, and discuss future enhancements such as multilingual support, user authentication, cloud deployment, and performance tracking.